Hannah Nguyen

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EDUCATION

Harvard College | GPA: 3.8/4.0 **|** Cambridge, MA **May 2027**

S.B. Candidate in Mechanical Engineering

- **Relevant Coursework**: Mechanics of Materials, Mechanical Vibrations, Thermodynamics, Computer Aided Machine Design, Intro to Electrical Engineering, Python for Engineers, Linear Algebra, Multivariate Calculus, Intro Computer Science
- **Activities**: Conflux Art Tech, MakeHarvard, Harvard Vietnamese Association, Asian American Dance Troupe

EXPERIENCE

Corning | Product Development Engineer Intern | Keller, TX **May 2026 - Present**

- Innovate and design optical connectors and adapters for fiber-optic telecommunications applications, collaborating closely with design engineers to develop and model high-precision components in Creo.
- Perform dimensional tolerance stack-up and statistical analyses to evaluate mechanical fit, optical alignment precision, and performance reliability; conduct reverse engineering to assess design elements and inform prototype development.
- Conduct patent searches and technical literature reviews to benchmark existing technologies and guide innovation; document design rationale, analytical findings, and prototype results through technical reports and formal engineering presentations.

IntuitiveMotion.ai | Mechanical Engineering Intern | Boston, MA **July 2025 - January 2026**

- Build and assemble mobile camera support structures using 80/20 extrusions, 3D-printed jigs, and hand-cut materials to achieve optimal field-of-view coverage; assisted in on-site troubleshooting and data collection for vision system development.
- Fabricate and rapid prototype mechanical components, including rotational and interlocking elements, using 3D-printed jigs, and hand-cut materials; tested multiple design variations to improve stability and performance.

Stephanie E. Pierce Lab | Biomechanics Research Intern | Cambridge, MA **Jan 2025 - August 2025**

- Design and fabricate biomimetic vertebral columns (BVCs) using 3Matic and custom 3D-printed molds; cast multipart silicone structures replicating early tetrapod spinal morphology for physical testing in aquatic environments.
- Use water-based flapping mechanisms to test BVC performance and evaluate trade-offs between stability and flexibility, analyzing dynamic metrics such as thrust, long-axis rotation, and cost of transport.

Aizenberg Laboratory | Research Intern | Allston, MA **Dec 2023 - Aug 2024**

- Prototype and demonstrate a proof of concept for an all-season window technology regulating indoor access to external cold.
- Rapidly develop acrylic devices and model houses to measure energy usage and determined optimal fluid concentrations in titanium dioxide and carbon water for specific temperature environments; Prototypes 20% more efficient than normal windows

PROJECTS

Bioinspired Launch Mechanism (Incorporated into Thesis) **Present**

- Engineer and optimize a quadrupedal launch mechanism for pterosaur-inspired (particularly the Quetzalcoatlus) robotics, iterating on latch-mediated spring systems using SolidWorks, COMSOL, and MATLAB.
- Rapid prototype and validate spring-loaded launch mechanisms with torsion springs, spring steel, 3D printing, and laser-cut components, refining structural integrity and launch efficiency through mechanical testing.

Bridge Building Competition **May 2025**

- Achieved a 1250:1 strength-to-weight ratio by designing and building a 0.2 kg all-wood arch bridge that withstood 250 kgf, using SolidWorks and FEA analysis to optimize truss geometry and reinforcement.
- Simulate structural performance to identify weak points; refine design with triangular bracing to maximize load capacity

SKILLS

Machining & Manufacturing: CNC Mill, Lathe, Bandsaw, Laser Cutter, 3D Printing, Silicone Casting, GD&T
Programming & CAD: SOLIDWORKS, Python, R, JavaScript, Arduino, MATLAB, MS Office, Creo, 3Matic, Trello